



B.Tech. Degree VI Semester Regular Examination April 2022

ME 19-205-0603 OPERATIONS MANAGEMENT (2019 Scheme)

Time: 3 Hours

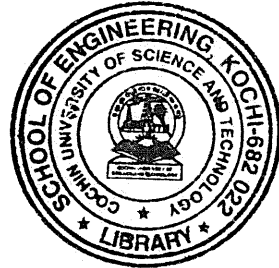
Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand different network techniques and forecasting methods.
 CO2: Learn production planning and control as well as inventory control techniques.
 CO3: Study the concepts of aggregate planning and scheduling.
 CO4: Get the basic knowledge of plant location and lay out and material handling.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze,
 L5 – Evaluate, L6 – Create
 PO – Programme Outcome



PART A (Answer **ALL** questions)

- | | | Marks | BL | CO | PO |
|-----|--|-------|----|----|----|
| I. | (8 × 3 = 24) | | | | |
| (a) | Sketch a dummy activity in network and mention its importance with respect to the succeeding activity. | 3 | L1 | 1 | 2 |
| (b) | What is the importance of demand forecasting in operations management? | 3 | L2 | 1 | 1 |
| (c) | Discuss the relationship between simplification, standardisation and interchangeability. | 3 | L4 | 2 | 2 |
| (d) | What is a bin card? Sketch and explain its use. | 3 | L2 | 2 | 2 |
| (e) | What is the role of aggregate planning in manufacturing operations? | 3 | L2 | 3 | 2 |
| (f) | What are the normal rules adopted for priority sequencing? | 3 | L1 | 3 | 1 |
| (g) | List out the significance of locating manufacturing unit in rural areas. | 3 | L1 | 4 | 2 |
| (h) | Differentiate between maintenance and replacement of machineries. | 3 | L4 | 4 | 2 |

PART B

(4 × 12 = 48)

- | | | | | | |
|-----------|--|---|----|---|---|
| II. | (a) Discuss the relationship of production function with other business activities of an Industry. | 6 | L3 | 1 | 2 |
| | (b) How forecasting methods are classified? Give examples for each. | 6 | L1 | 1 | 1 |
| OR | | | | | |
| III. | (a) Differentiate between any two project network analysis methods. | 3 | L2 | 1 | 2 |
| | (b) Construct the network diagram for the project given below. Also determine the critical path, expected project duration and probability of completing the project in 55 days. | 9 | L5 | 1 | 1 |

Activity	Immediate predecessor	Time (Days)		
		to	tm	tp
A	-	4	6	8
B	A	5	7	15
C	A	4	8	12
D	B	15	20	25
E	B	10	18	26
F	C	8	9	16
G	E	4	8	12
H	D,F	1	2	3
I	G,H	6	7	8

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IV.	(a)	Explain the steps involved in product design with the help of a flow chart.	6	L3	2	1
	(b)	What are the objectives of production planning and control? Discuss.	6	L2	2	2
OR						
V.	(a)	How industrial inventories are classified? Briefly discuss about VED and FSN analysis of Inventory management.	6	L2	2	1
	(b)	Monthly consumption of an item is 300 units. The price per unit is Rs.10. Inventory carrying cost is 18 per cent and ordering cost is 36 per order. Lead time of 1 month stock. Determine Re-order quantity, Re-order level, Minimum and maximum stock level, Average inventory.	6	L5	2	2
VI.		With a suitable example, explain the graphical and reaction rate methods of aggregate planning.	12	L4	3	3
OR						
VII.	(a)	What are the factors affecting scheduling decisions in manufacturing?	4	L1	3	1
	(b)	Discuss the procedure and conditions to be satisfied while performing sequencing of n Jobs and 3 machine problem with an example of your choice.	8	L4	3	2
VIII.	(a)	Explain the steps involved in plant layout design process.	6	L3	4	1
	(b)	List and explain the factors influencing plant location decisions.	6	L2	4	2
OR						
IX.	(a)	How material handling equipment's are classified? Explain with examples.	6	L1	4	2
	(b)	List and briefly explain the features of different maintenance methods (minimum 4 methods).	6	L2	4	2

Blooms's Taxonomy Levels

L1 = 20.83%, L2 = 30.00%, L3 = 15.00%, L4 = 21.66%, L5 = 12.50%
